WP-005 claim validation: seven topics checked

The paper is largely accurate but contains several errors requiring correction. Of the roughly 30 discrete factual claims examined, 20 are fully verified, 7 are partially verified or need nuance, and 3 are outright incorrect or outdated. The most consequential finding: the paper misses the formation of the Agentic AI Foundation (December 2025), the launch of agent marketplaces by all three hyperscalers, and the rebranding of LangGraph Platform to LangSmith Deployment — all developments that reshape the competitive landscape the paper describes.

Topic 1: LangGraph claims are solid with two corrections needed

GitHub stars: INCORRECT. The paper claims ~26,200 stars. The actual count on the `langchain-ai/langgraph` repository is **25.6k** as of mid-March 2026 (github) — an overstatement of roughly 600 stars (2.3%). Multiple independent snapshots from late February confirm the count was ~25,000 at that point, making 26,200 an unsupported figure. [practitioner/community evidence]

PyPI downloads: VERIFIED. pypistats.org reports **40,771,687 downloads** in the most recent month, closely matching the ~40 million claim. (PyPI Stats) The combined LangChain ecosystem exceeds 90 million monthly downloads. (Sequoia Capital) [vendor documentation]

Production deployments: ALL EIGHT VERIFIED, though evidence quality varies significantly:

- **Klarna** — Strongest evidence. Dedicated case study with CEO quote, quantified results (80% reduction in resolution time, 2.5M conversations). [vendor documentation + independent investor confirmation via IVP]
- **Elastic** — Independently confirmed on Elastic's own engineering blog, not just LangChain's materials. [vendor documentation from both parties]
- **Vodafone** — Two separate case studies (Vodafone corporate + Fastweb/Vodafone Italy's Super TOBi chatbot serving 9.5M customers). Named Vodafone senior manager quoted. [vendor documentation]
- **LinkedIn, Uber** — Confirmed via LangChain blog, conference presentations at Interrupt 2025, and Sequoia Capital's coverage. No independent case studies published by these companies. [vendor documentation]
- **J.P. Morgan, BlackRock** — Confirmed via Interrupt 2025 conference presentations (J.P. Morgan's "Ask David" investment research system; (Aibuilder) BlackRock's Aladdin Copilot platform). (TMCnet) These are public conference talks, not vendor marketing, which strengthens credibility. [practitioner/community evidence]

- **Replit** — Dedicated case study describing multi-agent architecture with manager, editor, and verifier agents. LangChain [vendor documentation]

Caveat: Most deployment evidence originates from LangChain's marketing materials. Elastic and Vodafone are the only companies with independent confirmation on their own domains.

LangSmith SOC 2 Type II and HIPAA: VERIFIED. LangChain's official changelog and documentation confirm both certifications. Langchain HIPAA BAAs are restricted to Enterprise plan customers. LangSmith is also GDPR compliant with EU data residency. LangChain [vendor documentation]

LangChain deprecation in favor of LangGraph: VERIFIED. The old `AgentExecutor`, `initialize_agent`, and `create_react_agent` (in `langgraph.prebuilt`) are all deprecated. LangChain 1.0's new `create_agent` is built directly on top of LangGraph's runtime. LangGraph is now the underlying engine for all agent work in the LangChain ecosystem. Medium [vendor documentation]

LangGraph v1.1.0: VERIFIED — released March 10, 2026, PyPI introducing opt-in `version="v2"` streaming format with full type safety, `GraphOutput` objects, and Pydantic auto-coercion. **LangGraph v1.1.2: FALSE — does not exist.** The latest release as of March 20, 2026, is v1.1.0. The v1.1.2 reference may be a confusion with `langchain-core` versions (which do have 1.1.x patch releases). [vendor documentation]

Topic 2: CrewAI figures hold up, but the Fortune 500 claim remains unverifiable

$18M funding: VERIFIED. Announced October 22, 2024, via GlobeNewswire. Led by Insight Partners (Series A) and boldstart ventures (inception round). SiliconANGLE Angel investors include Andrew Ng and Dharmesh Shah. CrewAI Confirmed by Tracxn, SiliconANGLE, and the investors themselves. [vendor documentation + practitioner/community evidence]

GitHub stars: VERIFIED. The repository shows **44.7k stars** as of March 2026, github consistent with the 44,700+ claim. Growth trajectory is well-documented: 34k (July 2025) Aibase → 40k (October 2025) CrewAI → 44.7k (March 2026). [practitioner/community evidence]

Named customers — all four verified:

- **PwC** — Selected CrewAI as a foundation for their "Agent OS" platform. PwC's Commercial CTO of GenAI (Jacob Wilson) quoted in materials. CrewAI [vendor documentation]
- **DocuSign** — Lead AI architect and lead data engineer presented at CrewAI's Signal Conference. Built a sales outreach system achieving **75% acceleration** in lead time-to-first-contact. Comparateur-IA [vendor documentation]

- **IBM** — Partnership announced October 2024, integrated with WatsonX.AI. (AIbase) [vendor documentation]
- **Konecta** — Independent BusinessWire press release (November 3, 2025) announces strategic alliance. Konecta established a dedicated CrewAI Centre of Excellence. (Business Wire) (Konecta) [vendor documentation + independent press release]

Fortune 500 60% claim: UNVERIFIABLE. The claim has escalated over time — from "nearly half" (October 2024) (Insight Partners) to "40%" (per eMarketer, sourcing CrewAI) to "60%+" (October 2025 onward). (CrewAI) All instances trace back to CrewAI's own statements. (Comparateur-IA) Founder João Moura acknowledges many companies "just don't want to be vocal," suggesting the figure likely counts any Fortune 500 employee who used the open-source framework, not enterprise contracts. (CrewAI) (Insight Partners) **No independent audit exists.** The paper correctly flags this as self-reported. [vendor documentation — self-reported]

New enterprise developments since the paper's likely research cutoff:

- **CrewAI AMP (Agent Management Platform)** launched October 2, 2025 (CrewAI) — positioned as "the first Agent Management Platform" (CrewAI) with RBAC, audit logs, agent training, and human-in-the-loop. (CrewAI +2) Naming has fluctuated between AMP and AOP. (CrewAI)
- **CrewAI OSS v1.0 GA** shipped October 20, 2025, with built-in observability. (CrewAI)
- **AMP Factory (on-premises)** supports private VPC deployment on AWS, Azure, or GCP. (CrewAI) (CrewAI) Claims SOC2 and FedRAMP High compliance paths. (Comparateur-IA)
- **Native MCP and A2A protocol support** added, with 12M+ daily agent executions claimed. (Nxcode) (Spaceo)
- Current open-source version is **v1.10.1** as of March 2026. (Nxcode)

Topic 3: Microsoft Agent Framework GA has not shipped yet

RC on February 19, 2026: VERIFIED. Official Microsoft Foundry Blog post by Shawn Henry confirms RC status for both .NET and Python on that date. (Microsoft) The GitHub releases page shows `python-1.0.0rc1` tagged February 19. (github) [vendor documentation]

GA by end of Q1 2026: NOT YET ACHIEVED. As of March 20, 2026, the latest release is `python-1.0.0rc3` (March 4, 2026) — still a release candidate. (github) In GitHub Discussion #4262, a Microsoft engineer stated GA is expected "around the end of March." (GitHub) The paper should note that **GA has not shipped as of the publication date** and the Q1 target may slip into early April. An important distinction: the **Foundry Agent Service** (a separate product) reached GA on

March 16, 2026, Dev|Journal but the Agent Framework SDK itself remains in RC. [vendor documentation]

Agent Governance Toolkit at 46 commits: VERIFIED. Direct inspection of `microsoft/agent-governance-toolkit` on GitHub confirms exactly 46 commits on main branch, 8 stars, 1 fork. github Latest release is v1.0.1 ("First Public Release," March 6, 2026). GitHub The toolkit covers all 10 OWASP Agentic Top 10 risks with **6,100+ tests**. GitHub It originated from a personal repository (`imran-siddique/agent-governance`) before migration to the Microsoft org. GitHub The low star count suggests early-stage adoption. [vendor documentation]

Agent 365 GA May 1, 2026: VERIFIED. Confirmed by the official Microsoft product page, Microsoft Microsoft Security Blog (March 9, 2026), and a CEO announcement from Judson Althoff. Microsoft Blogs Priced at **$15/user/month** standalone or included in Microsoft 365 E7 at $99/user/month. Rob Quickenden However, some security capabilities (runtime threat protection, investigation, hunting) will still be in **public preview** at GA — not all features launch fully on May 1. Samexpert [vendor documentation]

Topic 4: OpenAI Agents SDK version is newer than both figures

Latest version: v0.12.5, released March 19, 2026 — confirmed by PyPI with Sigstore cryptographic attestation. PyPI The paper's v0.12.3 is outdated, and the previous WP-004 research finding of v0.12.5 is correct. Release notes mention Responses API tool search support and gpt-5.4 model for computer use tool. GitHub [vendor documentation]

GitHub stars: APPROXIMATELY CORRECT. The repository (`openai/openai-agents-python`) displays **18.9k stars** on the main page, github with some GitHub CDN caches showing 19.3k. The 18,900 figure aligns with the 18.9k display and is reasonable as of March 2026. [practitioner/community evidence]

AgentKit: CONFIRMED as a real OpenAI product. This is not a confusion with another product. OpenAI launched AgentKit at **DevDay 2025 (October 6, 2025)** as "a complete set of tools for developers and enterprises to build, deploy, and optimize agents." It comprises three components: the **Agent Builder** (a visual canvas for creating multi-agent workflows with drag-and-drop nodes), the **Connector Registry** (centralized admin panel for data source management across OpenAI products), and **ChatKit** (embeddable chat UI toolkit). Full documentation exists at `developers.openai.com/api/docs/guides/agent-builder`. Note the name collision with Coinbase's separate "AgentKit" for crypto wallet interactions, which was itself built using OpenAI's Agents SDK. OpenAI [vendor documentation]

Topic 5: Marketplace commissions are accurate with one important nuance

GitHub Marketplace at 5%: VERIFIED. Official documentation confirms GitHub retains 5% since January 1, 2021 (reduced from the original 25%). No change through March 2026. GitHub [vendor documentation]

Salesforce AppExchange at 15%: VERIFIED, with the caveat that this applies to the standard **ISVforce** partnership model. OEM partners pay **25%** net revenue. Salesforce For ISVforce partners exceeding $20M in revenue share, the rate reportedly decreases. The 15% figure is correct for the most common partner type. [vendor documentation]

Atlassian Marketplace at 20-25%: PARTIALLY VERIFIED — requires significant update. Atlassian restructured its revenue share effective **January 1, 2026**. The 20-25% range now applies **only to Connect apps** (20% as of January 2026, rising to 25% on July 1, 2026). **Forge apps** — which Atlassian actively incentivizes — pay dramatically less: **0% for apps under $1M lifetime revenue**, and 16-17% for apps above that threshold. atlassian The paper's blanket "20-25%" statement is misleading without noting the Forge distinction, which represents Atlassian's strategic direction. [vendor documentation]

Topic 6: NIST initiative is real but the scope description needs tightening

The AI Agent Standards Initiative exists: VERIFIED. Officially launched **February 17, 2026**, by NIST's Center for AI Standards and Innovation (CAISI). nist Dedicated page at `nist.gov/caisi/ai-agent-standards-initiative`. Operates under three pillars: facilitating industry-led standards, fostering open-source protocol development, and investing in research on agent authentication and identity. nist [government source]

Timeline: VERIFIED with clarification. The April 2, 2026 deadline applies specifically to the **NCCoE concept paper "Accelerating the Adoption of Software and AI Agent Identity and Authorization"** — not a general initiative-wide comment period. A separate RFI on "AI Agent Security" had a March 9, 2026 deadline (already closed). Listening session registrations closed March 20, 2026. nist [government source]

Scope: PARTIALLY VERIFIED. Agent identity and authorization (roughly equivalent to "credentialing") are explicitly confirmed as core focus areas. However, **"privilege escalation safeguards" is not a term NIST uses.** The initiative addresses authorization, access controls, and security more broadly. nist The paper should use NIST's actual framing: "agent authentication, identity infrastructure, and authorization" nist rather than "privilege escalation safeguards." The initiative's scope is also broader than described, encompassing open-source protocol interoperability, international standards coordination (with EU's ENISA, Japan's AIST, ISO/IEC JTC 1/SC 42), and sector-specific adoption barriers. Meta Intelligence [government source]

Topic 7: Critical gaps the paper should address

The Agentic AI Foundation is the biggest omission

The **Agentic AI Foundation (AAIF)**, formed December 9, 2025 under the Linux Foundation, represents the single most significant organizational development in agent interoperability during the paper's coverage period. Linux Foundation PR Newswire Co-founded by Anthropic (donating MCP), OpenAI (contributing AGENTS.md), and Block (contributing goose), with platinum members including AWS, Google, Microsoft, Bloomberg, and Cloudflare, TechCrunch and gold members including Cisco, IBM, Oracle, Salesforce, SAP, Snowflake, Docker, and JetBrains. Linux Foundation This body governs the two emerging agent protocol standards — MCP and A2A — and its existence fundamentally changes the interoperability landscape. [vendor documentation]

A2A protocol has matured significantly

Google's A2A protocol has reached **v0.3** (July 31, 2025) with gRPC support and signed security cards. Apono It moved to Linux Foundation governance in June 2025. Linux Foundation Over **150 organizations** have joined, including all three hyperscalers: Microsoft integrated A2A into Azure AI Foundry and Copilot Studio with Entra Agent ID, AWS added native A2A support in Amazon Bedrock AgentCore, and Google embedded it in ADK and Agentspace. Enterprise implementations include Tyson Foods (supply chain), ServiceNow (AI Agent Fabric), SAP (Joule assistant), Adobe, and S&P Global. The critical framing: **MCP is "vertical" (agent-to-tool) while A2A is "horizontal" (agent-to-agent)** — they are complementary, not competing. Both are now under Linux Foundation governance. Let's Data Science CrewAI has added native A2A support; LangGraph has not. Openagents [vendor documentation]

All three hyperscalers launched agent marketplaces in 2025

This is a major gap in the paper's marketplace analysis:

- **Google Cloud AI Agent Marketplace** (April 2025) — uses A2A Agent Cards as standardized listing mechanism
- **AWS AI Agents and Tools Marketplace** (July 2025) — 900+ listings at launch, integrated with Bedrock AgentCore PYMNTS
- **Oracle Fusion Applications AI Agent Marketplace** (October 2025) — partner-built agent templates with a 21-point enterprise readiness checklist Oracle

These real-world agent marketplaces provide concrete benchmarks for the paper's theoretical revenue architecture. Agent Cards and templates are becoming the unit of distribution,

analogous to container images. Pricing models span free, subscription, usage-based, and outcome-based. [vendor documentation]

CrewAI Enterprise does claim immutable audit logs

The AWS Marketplace listing for CrewAI Enterprise Platform explicitly states: "CrewAI enforces role based access controls, policy driven approvals, and **immutable audit logs** for every agent execution." (AWS Marketplace) This narrows the governance differentiation the paper may be drawing against ContextKeeper. However, the "immutable" qualifier appears only in the AWS Marketplace listing, not prominently in CrewAI's own documentation or blog posts, which reference "RBAC + audit logs" without the immutability claim. The paper's competitive positioning should acknowledge this feature exists in CrewAI Enterprise while noting the limited documentation depth. [vendor documentation]

LangGraph Platform was rebranded to LangSmith Deployment

In October 2025, LangChain renamed LangGraph Platform to **LangSmith Deployment** and LangGraph Studio to **LangSmith Studio**, consolidating under three pillars: LangSmith Observability, LangSmith Evaluation, and LangSmith Deployment. (LangChain) Additionally, **Deep Agents** — a new open-source library built on LangGraph — launched in March 2026 with explicit planning tools, virtual filesystem, and subagent spawning capabilities. (MarkTechPost) (GitHub) LangSmith also added an **Agent Registry** for centralized agent management (GitHub) and an **Agent Builder** with skill creation and Slack integration. These features position LangChain as a full agent lifecycle management platform, not just an orchestration framework. [vendor documentation]

DAG-based agent pipelines are well-documented in research

Multiple peer-reviewed sources validate DAG-based execution as the dominant multi-agent orchestration pattern. **OrchDAG** (arXiv, October 2025) demonstrates DAG-based rewards for agent training. The **AOP framework** (VLDB CIDR'25) automates conversion from chain-format pipelines to DAG structures. A key production pattern has emerged: **hybrid architectures using classical DAG engines (Airflow, Dagster, Prefect) for scheduling and retries, with LangGraph/agent logic embedded within individual DAG tasks.** LangGraph natively supports both DAG and cyclic graph execution; (GitHub) (TrueFoundry) CrewAI achieves DAG-like behavior through task context chains (Markaicode) and `async_execution=True` but does not expose an explicit DAG mode. This validates the WP-005 architecture's DAG approach. [peer-reviewed + practitioner/community evidence]

New governance frameworks beyond the paper's coverage

- **ISO/IEC 42005:2025** (April 2025) — complementary standard for AI system impact assessments (AARC-360)

- **Airia AI Governance** (January 2026) — enterprise platform combining AI security, agent orchestration, and governance Airia
- **e& + IBM** (January 2026) — enterprise agentic AI governance using watsonx Orchestrate + watsonx.governance
- **Gartner prediction** that over 40% of agentic AI projects will be canceled by end of 2027 due to inadequate risk controls
- Only **14.4%** of organizations have AI agents go live with full security approval (Gravitee 2026 Report) Nemko Group AS

Summary correction table

Claim	Verdict	Action needed
LangGraph ~26,200 stars	**Incorrect** — actual is ~25,600	Correct to 25.6k
LangGraph ~40M PyPI downloads	**Verified**	None
8 production deployments	**All verified**	Note most evidence is vendor-sourced
LangSmith SOC 2 Type II + HIPAA	**Verified**	None
LangChain deprecated for LangGraph	**Verified**	None
LangGraph v1.1.2	**False** — does not exist	Correct to v1.1.0 (March 10, 2026)
CrewAI $18M October 2024	**Verified**	None
CrewAI 44,700+ stars	**Verified**	None
CrewAI named customers	**All four verified**	None
CrewAI Fortune 500 60%	**Unverifiable** — self-reported	Keep flagged
MS Agent Framework RC Feb 19	**Verified**	None

Claim	Verdict	Action needed
MS Agent Framework GA Q1 2026	**Not yet achieved**	Note still at RC3; may slip
Agent Governance Toolkit 46 commits	**Verified**	Note very early-stage (8 stars)
Agent 365 GA May 1, 2026	**Verified**	Note some features still in preview
OpenAI SDK v0.12.3	**Outdated** — now v0.12.5	Update version
OpenAI SDK 18,900 stars	**Approximately correct**	None
AgentKit as OpenAI product	**Verified**	None
GitHub Marketplace 5%	**Verified**	None
Salesforce AppExchange 15%	**Verified**	Note ISVforce-specific; OEM is 25%
Atlassian Marketplace 20-25%	**Partially verified**	Add Forge app rates (0-17%); note Jan 2026 restructuring
NIST AI Agent Standards Initiative	**Verified**	Tighten scope language; drop "privilege escalation"
April 2, 2026 comment period	**Verified**	Clarify it applies to NCCoE concept paper specifically

The paper's factual foundation is strong. The corrections above are mostly minor (star counts, version numbers). The larger concern is **missing context**: the AAIF formation, hyperscaler marketplace launches, LangSmith rebranding, and A2A protocol maturation all represent structural shifts that should be integrated into the competitive analysis and marketplace architecture sections.